

3GPP TS 38.321

Medium Access Control (MAC) protocol specification

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SYNTHETIC SAMPLE DOCUMENT — written for demonstration and testing of a retrieval-augmented chatbot. This is original content structured to resemble a 3GPP technical specification; it is NOT an official 3GPP publication and must not be used as a source of normative requirements.

5.1.2 Random access procedure

The random access procedure may be contention-based or contention-free. In the four-step contention-based procedure, the UE transmits a random access preamble (Msg1), receives a random access response (Msg2) granting uplink resources, transmits a scheduled message containing UE identity (Msg3), and finally receives a contention resolution message (Msg4). The two-step procedure combines the preamble and Msg3 payload into a single MsgA transmission to reduce latency.

5.4.2 HARQ operation

The MAC entity supports Hybrid Automatic Repeat reQuest (HARQ) with up to 16 parallel HARQ processes per serving cell for downlink and uplink respectively. Each HARQ process is identified by a HARQ process ID and maintains its own soft-combining buffer. Upon detecting a transport block error, the receiver shall request retransmission via HARQ feedback rather than waiting for RLC-layer retransmission, in order to minimise latency.

5.4.3 Logical channel prioritization

When the MAC entity is granted uplink resources, it allocates them to logical channels according to a prioritized list, respecting each logical channel's Prioritised Bit Rate (PBR) and Bucket Size Duration (BSD) before serving remaining channels strictly by priority order.

Annex A Modulation and coding scheme (MCS) index table (informative excerpt)

The table below is an illustrative excerpt of the MCS index to modulation-order mapping used for PDSCH/PUSCH scheduling decisions referenced elsewhere in this specification.

MCS index	Modulation order	Modulation scheme
0	2	QPSK
9	4	16QAM
16	6	64QAM
27	8	256QAM